Ishaan Kannan

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Education

Harvard University

Ph.D., Quantum Science and Engineering (Theoretical Physics focus) *2025 - present*

- Quantum Information and Computation, Machine Learning. GPA: 4.0/4.0
- Advisor: Jordan Cotler

California Institute of Technology

B.S., Physics (Major), Computer Science (Minor) *2020 - 2024*

- GPA: 4.2/4.3
- Research mentors: John Preskill, Hsin-Yuan (Robert) Huang, Leo Zhou

Publications

** denotes equal contribution. † denotes corresponding authorship on alphabetically-ordered work.*

I. Kannan*, S. Prabhu*, S.A. Khan, M. M. Sohoni, X. Song, S. Roy, A. Senanian, V. Fatemi, P.L. McMahon, J. Cotler. Exponential quantum advantage for learning signals with a single qubit, 2026. *arXiv: 2608.13521 [quant-ph]*. URL: https://arxiv.org/abs/2608.13521.

C.C.V. de Pradenne*, **I. Kannan***, H. Putterman, and J. Cotler. Restrictions on non-Clifford fault tolerance and ruling out beyond-SQL quantum metrology, 2026. *arXiv: 2607.27342 [quant-ph]* URL:https://arxiv.org/abs/2607.27342. Under review at *Physical Review Letters*.

N. Romanov, P. Ivashkov, Weiyuan Gong, **I. Kannan**, A. Gu, H. Hu, and S. Yelin. Learning Arbitrary Lindbladians with Quantum Error Correction, 2026. *arXiv: 2606.18188 [quant-ph]* URL:https://arxiv.org/abs/2606.18188. Under review at *PRX Quantum*.

I. Kannan*, H. Putterman*, and J. Cotler. Exponential speedups in fault-tolerant processing of quantum experiments, 2026. *arXiv: 2605.02057 [quant-ph]* URL:https://arxiv.org/abs/2605.02057. Under review at *Nature Communications*.

J. Cotler, D.L. Danielson, and **I. Kannan**†. Quantum Advantage for Sensing Properties of Classical Fields, 2026. *arXiv: 2602.17591 [quant-ph]*. To appear in *Physical Review X*.

J. Cotler, W. Gong, and **I. Kannan**†. Noisy Quantum Learning Theory. Nature Communications vol. 17, Article number: 6979 (2026). DOI: 10.1038/s41467-026-73693-x

I. Kannan, R. King, and L.Zhou. A Quantum Approximate Optimization Algorithm for Local Hamiltonian Problems, 2024. *arXiv: 2412.09221 [quant-ph]*. URL:https://arxiv.org/abs/2412.09221

H. Zhou*, L. Lewis*, **I. Kannan***, Y. Quek, H.-Y Huang, M.C. Caro. Learning Quantum States and Unitaries of Bounded Gate Complexity. PRX Quantum, vol. 5, p. 040 306, 4 Oct. 2024. DOI: 10.1103/PRXQuantum.5.040306

Awards and Honors

2026	**Hertz Fellowship, Finalist (50/1500)**, Hertz Foundation
	NDSEG Graduate Fellow (∼ 100/7000), US Department of Defense
	NSF Graduate Research Fellow (declined, 11 awards within subfield), National Science Foundation
2025	**QuICS Lanczos Graduate Fellowship (declined)**, University of Maryland
	Cover Feature Article, PRX Quantum
	International Year of Quantum Feature Collection, Selected Article, PRX Quantum
2024	**Richard P. Feynman Prize in Theoretical Physics**, California Institute of Technology
2021, 2022	**Summer Undergraduate Research Fellowship**, California Institute of Technology
2019	**National Merit Scholarship Finalist**, NMSC
2019	**Presidential Scholar Nominee**, US Dep. of Education

Presentations

Contributed Conference Presentations

Conference Talks

- Quantum Techniques in Machine Learning (QTML) 2024.
- Asian Quantum Information Sciences Conferece (AQIS) 2024
- Young Quantum Information Scientists Conference (YQIS). 2024.
- EPFL Workshop on Quantum Sensing, Learning, and Verification, Invited Talk, 2026.

Seminars

- Harvard Theory of Computation Seminar (2026)
- Harvard Quantum Information Seminar (2026)
- MIT Center for Theoretical Physics Quantum Information Seminar (2026)
- Caltech Summer Undergraduate Research Fellowship Seminar (2022)
- Caltech Summer Undergraduate Research Fellowship Seminar (2021)

Poster Presentations

- Annual conference on Quantum Information Processing (QIP) 2024.
- Theory of Quantum Computation conference (TQC) 2024

Teaching Experience

2023-24	**Quantum Learning Theory**, Student Instructor (*self-designed course*)	*Caltech*
	Graduate Quantum Mechanics, Teaching Assistant	
2022, 2023	**Electromagnetism and Special Relativity**, Teaching Assistant	*Caltech*
2021-24	**Dean's Tutor**, Physics, mathematics, and CS courses	*Caltech*

Other

Peer Review

Quantum Computing Theory in Practice (QCTiP) Conference, Berlin, 2025

Standardized Testing

SAT: 1590/1600

Extracurricular Activities

2020-24	**NCAA Div. 3 Tennis**, First singles, doubles player. National team rank 13 (school record). Western conferences individual rank 8. 2x First-Team All-SCIAC.	*Caltech*
2022-	**Pottery**, Amateur	
2008-	**Music**, 15 years violinist, 12 years pianist	